

WHITE PAPER

Small Engine ISAC

Information, licensing, and local production for a stronger propulsion industrial base

Prepared for Thomas Scott | Founding-member discussion | September 5, 2026

Executive summary

The Small Engine Information Sharing and Analysis Center would be a private, member-governed organization connecting propulsion developers, manufacturers, suppliers, customers, research institutions, and government partners. Its purpose is to turn fragmented industry information into better investment, product, licensing, and production decisions.

The center would combine engine and market intelligence with practical support for IP licensing, product licensing, and country-local manufacturing. It would connect technology owners with capable production partners, help participants navigate applicable national controls, and support qualification and sustainment of locally manufactured products. The intended outcomes are faster market access, stronger regional supply chains, protected intellectual property, and greater industrial capacity.

Information sharing and analysis are the organizing functions. Markets, manufacturing, technology, workforce, reliability, regulation, investment, and security all belong within the remit. Cybersecurity is one supporting domain, alongside physical and geopolitical risks, rather than the defining purpose.

This paper proposes a U.S.-based launch with international participation and initial emphasis on UAS propulsion. The engine scope includes reciprocating, rotary, turbine, electric, and hybrid systems. Scope boundaries, membership terms, and operating commitments would be adopted by the founding members.

Proposed launch decision

Recruit five founding organizations for a funded 90-day formation phase. Validate member demand, establish governance and information-sharing agreements, and build pilot engine and licensing directories. Expand to a professional operating team only after contracted recurring revenue supports the approved cost base.

Status: Independent concept proposal. All Small Engine ISAC dues, budgets, staffing, and milestones below are planning assumptions, not commitments. Existing organizations are benchmarks or prospective collaborators, not endorsers.

1. Industry need and institutional fit

The founding hypothesis is that useful propulsion technology, manufacturing capability, and customer demand are often separated by incomplete information and limited cross-border relationships. An engine developer may have a viable design but lack a production partner in a target country. A capable manufacturer may have available equipment but lack a qualified product. A vehicle developer may struggle to compare engines on a consistent installed-system basis. These are questions the founding phase should test directly with prospective members.

The opportunity is to provide shared analytical infrastructure that individual firms would otherwise assemble repeatedly: comparable product records, verified capability profiles, licensing availability, regulatory pathways, and evidence on production readiness. The ISAC would make that information more useful through expert review, trusted introductions, and member-led analysis.

Relationship to existing ISACs

The National Council of ISACs describes a remit spanning cyber and physical threats and other hazards. The proposed center would build on that information-sharing model and add a propulsion-specific economic and industrial-development mission. That broader mission is a design choice, not a claim of existing government designation or National Council membership. [1]

Organization	Published focus	Potential relationship
Space ISAC	Space-sector intelligence and collaboration	Operating and membership benchmark
Aviation ISAC	Aviation cybersecurity and supply chains	Aviation threat-information partner
Manufacturing ISAC	Manufacturing cybersecurity and resilience	Manufacturing risk-information partner

Sources: Official organizational pages. [2–4] Public-source review has not identified an exact small-engine counterpart; this is not proof that no private group or working group exists.

Initial technical boundary

Launch around propulsion for unmanned aircraft and closely related applications. Include reciprocating and rotary engines, small gas turbines, electric propulsion, hybrid power systems, controls, fuel systems, generators, gearboxes, thermal management, critical components, and testing. Admit adjacent applications where they share suppliers, production processes, or technology. Set any numerical power or thrust limits after reviewing the founding cohort rather than treating “small” as a settled technical definition.

Maintain thrust in newtons for direct-thrust systems and shaft or electrical power in kilowatts for power-delivery systems. Compare fuel or energy consumption, duty cycle, conditions, and installed mass explicitly. Any conversion between thrust and power requires stated flight and efficiency assumptions.

2. Member value and core services

Market and technology intelligence

A curated engine atlas would link developers, product families, technical specifications, maturity, installed-system boundaries, and production locations. Demand assessments would distinguish funded orders, announced programs, and scenario assumptions. Technology roadmaps would identify common performance gaps and opportunities for precompetitive research without prescribing individual company product strategy.

IP, product licensing, and local production

A licensing exchange would publish owner-approved descriptions of available IP and products, desired territories, partnership needs, and manufacturing requirements. Qualified introductions would connect rights holders with prospective licensees. Country-specific production pathways would bring together commercial rights, technology-transfer permissions, local sourcing, quality assurance, and market-entry requirements.

Manufacturing capacity and supplier development

Capability profiles would describe processes, equipment, certifications, production experience, test assets, and workforce skills. Capacity analysis would distinguish installed capacity, demonstrated output, and owner-reported availability. Shared qualification resources and second-source opportunities would help members convert available capability into production readiness.

Reliability, regulation, and resilience

Members could contribute anonymized field findings, maintenance lessons, manufacturing escapes, and component obsolescence notices. Analysts would connect those observations with supply-chain exposure, changing national controls, standards activity, and physical or cyber disruptions. Findings would retain operating context and evidence quality so that members can judge their relevance.

Investment and partnership intelligence

The center would map technology gaps, development milestones, licensable assets, and owner-authorized partnership or acquisition interest. Investors and industrial partners would gain a more consistent basis for screening opportunities. Independent experts could support company-specific diligence through separately scoped engagements.

Member group	Primary benefit
Developers and IP owners	More qualified licensees and routes to market
Manufacturers and suppliers	Product opportunities and qualification pathways
Vehicle developers and customers	Comparable propulsion options and supply visibility

Member group	Primary benefit
Research and test organizations	Industry requirements and utilization opportunities
Investors and government customers	Better evidence for industrial-base decisions

3. Licensing and country-local production

Licensing would be a core economic function. The ISAC would support foreign designs entering U.S. production, U.S. designs entering partner-country production, and partnerships between non-U.S. countries. The benefit is to connect proven technology with local industrial capability while protecting the commercial interests of developers and manufacturers.

Commercial rights and regulatory pathways

Commercial licensing addresses patents, know-how, software, product designs, manufacturing rights, territories, royalties, sublicensing, and improvement ownership. Regulatory authorizations address the particular technology, services, participants, destinations, and activities involved. The ISAC would help members organize both workstreams early, supported by qualified counsel and relevant authorities.

For U.S.-controlled defense technology, DDTC agreement mechanisms include Technical Assistance Agreements and Manufacturing License Agreements. BIS administers EAR requirements for covered goods, software, and technology, including relevant reexports and transfers. UK and EU regimes provide additional country and regional requirements. Country dossiers would identify applicable pathways and record continuing conditions, enabling reliable production and market-access plans. [5–8]

Proposed opportunity-to-production workflow

Discovery: An owner-approved opportunity profile identifies the product, maturity, available rights, preferred markets, and minimum licensee capabilities. Shared listings begin with information cleared for the intended audience.

Partner qualification: The center screens manufacturing experience, financial capacity, quality systems, ownership information, test resources, and the proposed market. Detailed commercial and technical diligence moves into a permissioned project workspace.

Agreement development: The parties and their advisers establish commercial rights, background IP, improvement rights, technical-support obligations, relevant authorizations, and responsibilities for configuration control. The ISAC provides frameworks and coordination; the parties approve their agreements.

Localization and qualification: The partners define local and imported content, supplier qualification, tooling, training, process validation, acceptance tests, and customer requirements. Local-content and procurement eligibility are assessed for the actual program and production configuration.

Production and sustainment: The licensee and rights holder maintain approved configurations, change notices, quality records, support arrangements, and renewal obligations. Consented, aggregated lessons improve the center’s future guidance.

Illustrative use case

A European rotary-engine developer seeks access to a U.S. vehicle program. The ISAC identifies an interested U.S. producer, maps process and test gaps, and connects the parties with licensing and export-control expertise. The partners define a U.S. production configuration and qualification plan. The measurable benefit is reduced time from partner discovery to a qualified local source. This is an illustrative workflow, not a claimed transaction or eligibility determination.

4. Information products and trusted participation

Product	Initial cadence	Decision supported
Engine and technology atlas	Quarterly refresh	Product screening and comparison
Licensing opportunity register	Rolling owner updates	Partner and territory matching
Manufacturing and test directory	Quarterly validation	Production and qualification planning
Market and policy briefing	Monthly	Demand and market-entry planning
Industrial-base assessment	Annual	Investment and capacity priorities
Disruption and reliability notices	Event-driven	Operational response and prevention

Proposed service levels. Publish correction notices when material information changes and display the most recent verification date on every record.

Evidence quality

Every record should include source, date, responsible organization, permitted use, and confidence status. Distinguish published specifications, member assertions, reviewed documents, and independently observed test results. Engine records should show whether mass includes controls, cooling, exhaust, generator, gearbox, or energy storage. Production location and corporate nationality should be separate fields. Product counts should never be presented as production share or market share.

Participation and access

Use three access layers: public information; member-approved shared information; and project-specific workspaces. Contributors retain ownership and authorize defined uses. International access is governed by the information and project permissions involved, with country-specific collaboration groups where useful. Membership level does not override a contributor’s permissions.

A lightweight launch platform would combine a searchable directory, document repository, member authentication, access logs, and a correction process. Sensitive design packages should remain with their owners until a project workspace and its access requirements are approved. The center's initial investment should favor data quality and analyst capacity over custom software development.

Member contribution model

Ask each industrial member to nominate one technical contact and one commercial contact, validate its own profile quarterly, and contribute to at least one working group. Demonstrated usefulness should drive participation: more accurate records attract better inquiries, while shared lessons reduce repeated effort across the membership.

5. Governance and operating structure

Establish a private, member-governed organization with an independent executive director and a balanced board. A proposed seven-seat board would include representation from developers, manufacturers, suppliers or customers, and an independent chair. Reserve meaningful representation for small companies. Founding status should recognize formation support while board terms and elections maintain accountability.

Neutrality and member confidence

Adopt transparent rules for research priorities, partner referrals, sponsorship, and conflicts of interest. Company-specific negotiations remain with the participating firms. An approved competition-compliance policy, facilitated meetings, documented agendas, and aggregated reporting support constructive collaboration among competitors. Member-approved disclosure and access agreements enable useful exchange while preserving each participant's commercial position.

Legal form and service delivery

A U.S. nonprofit business league is one structure to evaluate, not an assumed tax status. IRS guidance distinguishes improvement of conditions across a line of business from services primarily performed for individual firms. Counsel should assess the actual activity mix before selecting a 501(c)(6) structure. Industry-wide research, education, and common frameworks would form the shared core; bespoke transaction work could be delivered by independent providers or an appropriately structured separate service operation. [9]

Initial operating team

An executive director would own strategy, membership, partnerships, and financial management. An industry analyst would maintain evidence quality and analytical products. A membership and licensing program manager would coordinate contributions, introductions, and localization workstreams. Contract specialists would provide IP, export-control, accounting, and other expertise as required. Hiring would follow committed revenue, beginning with fractional support during formation.

Working groups and external relationships

Launch three working groups: licensing and localization; technology and manufacturing; and market intelligence and resilience. Reliability, workforce, regulatory, and security topics would sit within these groups until activity justifies dedicated teams. Government agencies, laboratories, and existing ISACs could participate through advisory or project relationships. Outreach would explore mutual value without assuming endorsement, funding, or membership commitments.

The center would publish an annual activity and financial report for members, with a dashboard covering engagement, data quality, supported partnerships, and operating sustainability. Sponsorship recognition and board representation would remain separate from analytical findings.

6. Membership economics

Published ISAC dues provide price references, not proof of willingness to pay for this new center. Space ISAC lists corporate Silver, Gold, and Platinum memberships at \$10,000, \$25,000, and \$50,000 annually. Small-business levels are \$2,500 and \$5,500, with revenue eligibility; university and laboratory levels are \$4,500 and \$5,500. IT-ISAC publishes annual dues starting at \$3,000. These benchmarks were reviewed September 5, 2026. [2,10]

Proposed annual dues for validation

Category	Eligibility / role	Annual dues
Emerging company	Annual revenue below \$5M	\$2,500
Small company	\$5M to below \$25M	\$5,000
Growth company	\$25M to below \$100M	\$10,000
Industry member	\$100M to below \$1B	\$20,000
Strategic member	\$1B or more	\$35,000
Founding member	Formation commitment; replaces base dues	\$50,000
University / laboratory	Research participation	\$2,500 / \$5,000

Proposed amounts, not adopted prices. Revenue is USD-equivalent consolidated revenue; founding members would approve affiliate treatment and currency-conversion rules. Each organization pays one category, with no double counting.

Membership value and supplementary revenue

All industrial tiers would receive core intelligence, an organizational profile, licensing listings, qualified introductions, and working-group participation. Higher tiers could add user accounts, event passes, and expanded analyst access. Sensitive information remains governed by permissions. This preserves access for smaller developers whose technology may be strategically important.

Separate sponsorships, events, commissioned studies, and scoped services could support additional work. Government agencies may participate as nonvoting advisers and purchase defined analytical services when authorized. Neither sponsorship nor public funding is assumed in the base model. No currently verified ISAC-formation award is identified in this paper; any later application must be matched to a specific solicitation and eligible deliverable.

7. Illustrative operating model

The scenario below is a steady-state annual run rate, not a forecast of first-year collections. It assumes all listed members pay for a full year. Recruitment timing, discounts, nonpayment, and attrition would reduce realized cash receipts.

Membership mix	Count	Dues each	Annual revenue
Founding	5	\$50,000	\$250,000
Industry	8	\$20,000	\$160,000
Growth	10	\$10,000	\$100,000
Small	15	\$5,000	\$75,000
Emerging	10	\$2,500	\$25,000
Universities	3	\$2,500	\$7,500
Laboratories	2	\$5,000	\$10,000
Total	53		\$627,500

Illustrative recurring dues only. Founding dues replace, rather than supplement, the member's revenue-based tier.

Annual operating allocation

Expense assumption	Annual amount
Executive director, salary and employer costs	\$220,000
Industry analyst, salary and employer costs	\$120,000
Membership/licensing manager, salary and employer costs	\$110,000
Contract legal, IP, and export-control expertise	\$45,000
Platform, data, and IT support	\$30,000
Accounting, insurance, and administration	\$25,000

Expense assumption	Annual amount
Member meetings and travel	\$25,000
Total operating expenses	\$575,000
Unallocated operating balance	\$52,500

Planning allocations, not compensation benchmarks or supplier quotes. Employer costs include payroll burden and benefits. One-time formation costs require a separate approved budget.

At \$575,000 per year, average monthly operating cost is approximately \$47,917. A three-month operating reserve is \$143,750. Losing two \$50,000 founding renewals would reduce annual revenue to \$527,500, creating a \$47,500 shortfall against this cost base. Formation commitments should cover startup work and reserves; recurring staffing should follow collected or reliable contracted revenue. Review founding renewals and revenue concentration quarterly.

8. Launch roadmap and success measures

Days 1–30: Confirm demand and founding commitments

Conduct 15–20 structured interviews across developers, manufacturers, vehicle customers, and research organizations. Test the value of the three initial products: engine atlas, licensing register, and manufacturing directory. Seek five founding commitments, nominate the organizing board, and commission governance and legal design. Interview counts and commitments are proposed targets, not completed activity.

Days 31–60: Build a usable pilot

Adopt the charter, data rules, contribution agreements, and service scope. Establish a basic member platform and populate at least 30 source-traceable engine records, 10 owner-approved licensing opportunities, and 15 manufacturing or test profiles. Review sample records with members and record verification status and correction turnaround.

Days 61–90: Demonstrate member benefit

Deliver the first market and capability briefing, run a licensing and localization workshop, and complete five qualified partner introductions. Obtain specific member feedback on usefulness and willingness to renew. Approve the full-year operating plan only when recurring revenue and a cash-flow forecast support the expense commitments and reserve policy.

Months 4–12: Establish recurring value

Publish monthly briefings and the first industrial-base assessment, broaden verified coverage, and develop country-specific localization dossiers for member-selected markets. Aim to support three bilateral feasibility or diligence workstreams and one documented qualification or localization plan.

These are activity targets; signed licenses, production awards, and approvals remain outcomes controlled by the participating parties.

Measure	How to assess
Member economics	Paid renewals, recurring revenue, cash runway
Information quality	Source coverage, freshness, resolved corrections
Licensing usefulness	Qualified introductions progressing to diligence
Production impact	Documented qualification and localization milestones
Decision value	Member-confirmed time saved or decisions improved

Founding-member proposition

The Small Engine ISAC would give members a shared view of propulsion technology and industrial capability, then help convert that knowledge into productive partnerships. Its distinguishing contribution is the connection between analysis, IP and product licensing, and country-local production. Founding members are invited to validate this proposition, define the initial services, and fund a disciplined launch that produces measurable industrial and commercial value.

Sources and evidence notes

Official public sources reviewed September 5, 2026. Numbered references support the nearby factual statements. Proposed services, governance, dues, financial scenarios, and launch targets are original planning recommendations. The paper does not quantify market size or domestic market share because a consistent, validated dataset has not yet been established. No affiliation with Re:Build or any named organization is implied.

[1] National Council of ISACs. Definition, hazards covered, and sector collaboration.

<https://www.nationalisacs.org/>

[2] Space ISAC, Membership. Membership model and published annual dues.

<https://spaceisac.org/membership/>

[3] Aviation ISAC. Published aviation and supply-chain focus.

<https://www.a-isac.com/>

[4] Manufacturing ISAC. Published manufacturing focus.

<https://www.mfgisac.org/>

[5] U.S. Department of State, DDTC, Agreement Guidance. Technical Assistance and Manufacturing License Agreements.

https://www.pmdtdc.state.gov/deccs?id=ddtc_kb_article_page&sys_id=KB0010111

[6] Bureau of Industry and Security, Determine what is subject to the EAR. Jurisdiction and covered exports, reexports, and transfers.

<https://www.bis.gov/licensing/determine-what-is-subject-to-the-EAR>

[7] UK Government, Export controls: military goods, software and technology. UK technology-transfer requirements.

<https://www.gov.uk/guidance/export-controls-military-goods-software-and-technology>

[8] European Commission, Exporting dual-use items. EU dual-use framework and technical assistance.

https://policy.trade.ec.europa.eu/help-exporters-and-importers/exporting-dual-use-items_en

[9] Internal Revenue Service, Business leagues. Common-interest purpose and limits on particular services.

<https://www.irs.gov/charities-non-profits/other-non-profits/business-leagues>

[10] IT-ISAC, Who we are. Published membership entry price.

<https://www.it-isac.org/who-we-are>